

Diego Aguillon

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EDUCATION

The University of Texas at Austin

Bachelor of Science in Computer Science, Minor in Government

Coursework: Operating Systems, Algorithms, Machine Learning I, iOS Mobile Computing, Computer Vision

May 2027

GPA: 3.95

TECHNICAL SKILLS

- **Development:** Java, C, C++, Swift, Python, Linux, ARM64, x86, HTML/CSS
- **Tools / Software:** Git, Visual Studio Code, Xcode, Eclipse IDE, LaTeX

PROJECTS

Generative AI Partner

July 2025 – August 2025

- Developed Python module that extracts public financial data on 10K+ companies from SEC's EDGAR system
- Automated daily extraction and parsing of 500K+ fiscal records with AWS Lambda and Amazon EventBridge
- Integrated with Amazon Bedrock API to train Claude AI on yearly (10-K) and quarterly (10-Q) financial reviews

Virtual Operating System Kernel

January 2025 – May 2025

- Built preemptive kernel for multi-threaded operating system in C++ and x86 assembly via QEMU emulator
- Implemented Ext2 file system capable of traversing directories and interpreting files and symbolic links
- Included virtual memory functionality to enable construction and maintenance of user processes
- Simulated PS/2 keyboard and mouse drivers allowing for interpretation of user input

Computer Instruction Pipeline Emulator

November 2024 – December 2024

- Simulated five-stage central processing unit pipeline in C and ARM64 assembly
- Emulated critical hardware including arithmetic logic unit and general purpose registers
- Implemented core functionality that fetches, decodes, and executes computer instructions
- Increased throughput and processor efficiency via integration of instruction forwarding and hazard control

WORK EXPERIENCE

Data Structures Undergraduate Course Assistant

August 2024 – Present

The University of Texas at Austin, Austin, TX

- Offer live walkthroughs and debugging of data structures problems in Java for 4 office hours a week
- Direct weekly hour-long discussions with 20+ students regarding course content and practice problems
- Provide instructional feedback on assignments for 6+ hours a week via discussion board and grading interface

Computer Science Undergraduate Advising Office Associate

August 2024 – Present

The University of Texas at Austin, Austin, TX

- Field phone calls and assist walk-ins with computer science advising needs for 5+ hours a week
- Manage drop-in appointments and welcome events for 14+ hours a week during summer orientations

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Career Prep Fellow

January 2025 – Present

Management Leadership for Tomorrow

- Accepted into a selective 18-month professional development program for high-achieving talent
- Complete business case studies and assignments to grow leadership and technical skills
- Attend conferences hosted by industry leaders such as Apple, Google, and Dell Technologies

Director of Logistics

January 2024 – Present

Information and Systems Security Society, Austin, TX

- Develop and enforce yearly budget for \$20K worth of organization assets as director of logistical operations
- Facilitate and execute 15+ weekly talks, workshops, and Capture the Flag events for 100+ members
- Coordinate with executive board to manage company sponsor tiers and plan 5+ sponsored events per semester
- Manage promotion, outreach, and financial logistics for 1000+ participating teams in annual, international UTCTF